

Improved Model for the Intensity of Low-Energy Tailing in Si(Li) X-Ray Spectra

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Accurate tail-to-peak intensity ratios are extracted from the spectra produced in a Si(Li) detector by x-rays of energies 2–10 keV. The widely used model of a surface layer of incomplete charge collection fails to explain the energy dependence of these data. The model is augmented to include loss of photoelectrons travelling back into the ICC layer from an interaction in the active silicon region; this gives excellent agreement with experiment. It is shown that the detection efficiency in the region of low x-ray energy may be deduced from the tail-to-peak ratios, together with measurement of the thicknesses of the beryllium window and the metal contact; a potential source of error in measuring the latter by the popular fluorescence technique is demonstrated.

INTRODUCTION

The concept of a surface layer of incomplete charge collection (ICC) in Si(Li) detectors is widely used^{1,2} in determining the efficiency of these devices at low x-ray energies. The objective of this work was to investigate the effective thickness of this layer through measurements of the detector's response to monoenergetic x-rays and thence to test different models which seek to explain in physical terms the loss of ionization charge in this region. An improved model is suggested and its ramifications for efficiency determination are explored.

MODELS FOR SURFACE LAYERS

The response function of a Si(Li) x-ray detector, shown schematically in Fig. 1, is basically a Gaussian profile accompanied by a low-energy continuum or 'tail' extending down to zero energy in the pulse-height spectrum. This tail is usually attributed in very general terms to escape of photoelectrons or to loss of ionization electrons via poorly understood processes such as recombination, trapping and diffusion to the surface. Often a distinction is made between a very thin surface dead layer which is completely inert and a subsequent much thicker ICC layer (as in Fig. 2). Interaction of x-rays within the former would result in fluorescence of silicon K α x-rays, about half of which would then be detected, resulting in a peak in the spectrum at 1.74 keV. Recent measurements to this end^{1–3} have revealed no such peak and have placed upper limits ranging from 0.01 to 0.04 μm on its thickness.

The ICC layer is generally taken to be responsible for the low-energy tail. The first discussions of such a layer were those of Goulding⁴ and Llacer *et al.*⁵ If all of the tailing is attributed to interactions in this layer, or conversely if every x-ray interaction in the layer results in a degraded event, then the layer thickness t_{ICC} can be cal-

culated from the equation

$$\frac{A_T}{A_G} = \frac{1 - \exp(-\mu^* t_{\text{ICC}})}{\exp(-\mu t_{\text{ICC}})} \quad (1)$$

where μ^* is the linear attenuation coefficient of silicon for the relevant x-ray energy, A_T is the integrated intensity of low-energy tailing events and A_G is the integrated Gaussian peak intensity, as indicated in Fig. 1. Llacer *et al.*⁵ obtained a value of 0.15 μm by measuring the tail-to-total ratio for low-energy x-rays. Using fluorescent K x-rays of energy 1.3–3.7 keV, Shima *et al.*⁶ deduced t_{ICC} values of 0.05, 0.1 and 0.2 μm for three detectors; the deduced value was stated to be essentially

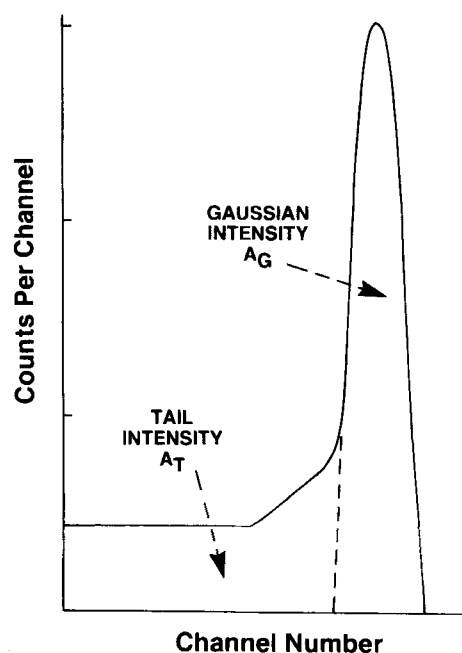


Figure 1. X-ray peak with integrated counts (intensity) A_G in the Gaussian portion and A_T in the tailing portion.

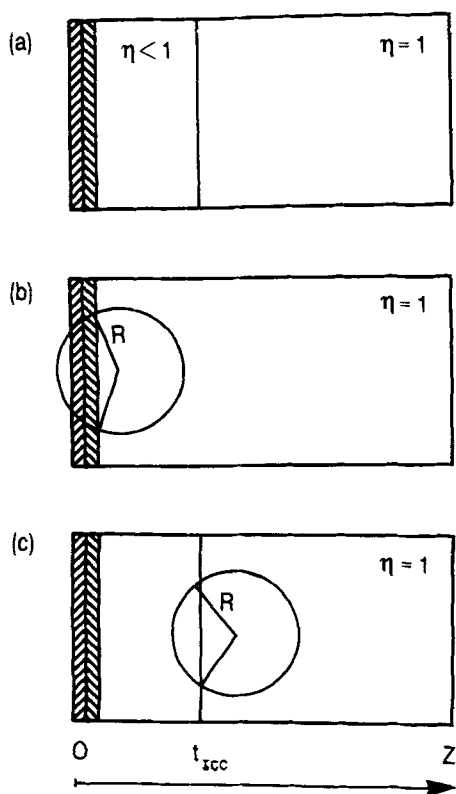


Figure 2. Models of the front region of a Si(Li) x-ray detector. The shaded layers at the left represent the metal contact and the surface dead layer in which $\eta = 0$. In (a) the tail imputed to an ICC region of $\eta < 1$. In (b) there is no ICC region and tailing is imputed only to escape of photoelectrons and Auger electrons from the depleted region where $\eta = 1.0$. The model (c) proposed here includes an ICC region, and in addition the escape into this region of photoelectrons from the $\eta = 1$ region.

independent of x-ray energy. Similar approaches have been made by other workers. In addition, various workers have deduced the value of t_{ICC} via model fits to the measured efficiency–energy relationship.² Baker *et al.*⁷ described an elegant means of determining t_{ICC} via the intensity ratio of full-energy peaks when an x-ray beam enters the detector at two widely different angles.

For small thicknesses, Eqn (1) reduces to

$$A_{\text{T}}/A_{\text{TG}} = \mu^* t_{\text{ICC}} \quad (2)$$

where A_{TG} is the combined intensity of tail and Gaussian. Nicholson *et al.*,⁸ using fluorescent x-rays between 1 and 6 keV, presented data to show that indeed the ratio of tail to total counts depends linearly on the mass attenuation coefficient μ . However, their plot of $A_{\text{T}}/A_{\text{TG}}$ versus μ has a non-zero intercept which is not predicted by the ICC model, and so they cited two separate mechanisms as causing the tailing. From the slope of the linearly dependent portion they obtained an ICC layer thickness of 0.47 μm ; they did not address the origin of the other component. Geretschlager³ recently studied two detectors using proton-induced K x-rays in the energy range 1–9 keV. He showed that assignment of the entire tail to the ICC layer results in t_{ICC} values that are not unique but have a well defined dependence on x-ray energy. He therefore followed Nicholson *et al.*

in postulating two mechanisms so that only one component arises from an ICC layer of well defined thickness; he explored the other via Monte Carlo calculations that simulate the effect of clusters of imperfections within the main portion of the crystal.

An entirely different approach is that of Shima *et al.*,⁶ who calculated tail-to-peak ratios due to escape from the sensitive volume of photoelectrons and Auger electrons, in the model in Fig. 2(b). This differs from the ICC layer model [Fig. 2(a)] by assuming two regions of zero and 100% charge collection efficiency, and then calculating the escape of electrons from the 100% region back into the zero region as a function of photon interaction depth (assuming straight paths and using Fitting's range equation⁹ for the electrons). Thus all the tailing is imputed to unavoidable physical processes rather than undefined deficiencies of charge collection. For two of Shima *et al.*'s three detectors the prediction was about five times less than the observed tail-to-peak ratio; for a third there was agreement in the vicinity of 3 keV energy but the energy dependence was markedly different. For the detector recently studied by Krumrey *et al.*,¹⁰ Shima *et al.*'s model again predicts too low a tailing intensity. These observations suggests that in all cases studied there are sources of charge loss (as implied in the ICC model) in addition to the unavoidable effects of escape of photoelectrons and Auger electrons (Shima *et al.*'s model).

EXPERIMENTAL

The Si(Li) detector used was a Series E device manufactured by Link Analytical, chosen on account of its very low tailing intensity. Its energy resolution is 126 eV [full width at half maximum intensity (FWHM)] at 5.9 keV. Its active surface area is defined by an internal 5-mm diameter tantalum collimator. Monoenergetic x-ray beams were obtained from a curved crystal monochromator as described previously.¹¹ Improved design extended the energy range covered to 1.7–20 keV; here we used $\text{L}\alpha_1$ (Zr, Mo, Rh, Ag) and $\text{K}\alpha_1$ (Si, Ti, Cr, Fe, Ni, Cu, Ge) to cover the energy range 1.7–12 keV. With the Si(Li) detector placed just outside the monochromator exit window (0.05-mm Mylar), spectra were accumulated at 1000 s^{-1} to an intensity of 2×10^6 counts.

In a preliminary experiment an iron $\text{K}\alpha_1$ x-ray beam was collimated to 0.4-mm diameter and the detector was stepped across the beam. The resulting radial dependence of $A_{\text{T}}/A_{\text{G}}$ is shown in Fig. 3. In order to eliminate the obvious edge effects, a 3-mm diameter collimator was attached to the detector in all subsequent work, during which several spectra were recorded using all the targets indicated.

The spectra were found to be well described by the fitting function

$$F(i) = G(i) + D(i) + S(i) + S_{\text{T}}(i) + G_{\text{e}}(i) \quad (3)$$

where the components are as follows:

$$\text{Gaussian:} \quad G(i) = H_{\text{G}} \exp \left[\frac{-(i - i_0)^2}{2\sigma^2} \right] \quad (4)$$

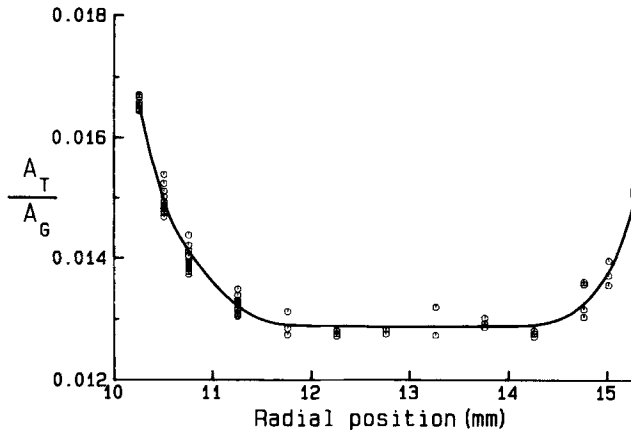


Figure 3. Radial dependence of the overall tailing intensity for 6.404-keV x-rays.

Exponential:
$$D(i) = \frac{1}{2}H_D \exp\left(\frac{i-i_0}{\beta}\right) \times \operatorname{erfc}\left(\frac{i-i_0}{\sigma\sqrt{2}} + \frac{\sigma}{\beta\sqrt{2}}\right) \quad (5)$$

Shelf:
$$S(i) = \frac{1}{2}H_S \operatorname{erfc}\left(\frac{i-i_0}{\sigma\sqrt{2}}\right) \quad (6)$$

Truncated shelf:
$$S_T(i) = \frac{1}{2}H_{ST} \left[\operatorname{erfc}\left(\frac{i-i_0}{\sigma\sqrt{2}}\right) - \operatorname{erfc}\left(\frac{i-i_T}{\sigma\sqrt{2}}\right) \right] \quad (7)$$

Escape peak:
$$G_e(i) = H_e \exp\left[-\frac{(i-i_e)^2}{2\sigma^2}\right] \quad (8)$$

Background:
$$B(i) = a + bi \quad (9)$$

where i represents the channel number, i_0 is the Gaussian centroid, i_e the escape peak centroid and i_T the channel where the truncated shelf has fallen to half its height. The H parameters are the heights of the various components and determine the slope of the exponential tail.

The main Gaussian portion represents the broadening effect of noise and of the statistics of charge formation. The exponential and shelf features are the widely accepted empirical representation of the effects of incomplete charge collection arising from detector imperfections; to date it has not been possible to identify either of these in an unambiguous manner with particular processes. The truncated shelf is another empiricism introduced for the first time here in response to the observation of such a phenomenon in the spectra. The complementary error functions in these terms arises from the convolution of simple flat shelf and exponential terms with a unit-area Gaussian to obviate non-physical sharp edges at $i = i_0$ and i_T .

The 'tailing' terms are not required to account for Compton scattering in the housing. The effect of such scattering is evident when the incident beam is not well collimated, but in the measurements described here a

parallel beam was directed accurately onto the central portion of the front face of the detector, and no photons were incident on the housing. Forward scattering in the dead layer or window appears to be a negligible contributor given the relevant thicknesses and the Compton scattering cross-sections.

The escape peak arises from escape of silicon K x-rays from interactions near the front surface. Finally, the environmental background is observed to be linear (essentially horizontal) during long runs with no x-ray source present.

The total intensity A_T in the tail components $D(i)$, $S(i)$ and $S_T(i)$ was obtained directly from the fits.

RESULTS AND DISCUSSION

Figure 4 shows several measured lines with the fits of Eqn (3) superimposed. Attention should be drawn to the fact that the shelf component $S(x)$ is a good representation from the peak energy essentially all the way down to the cut-off imposed by the lower level discriminator of the ADC. In our previous work on other detectors a good fit was obtainable only in a fairly narrow region of fit excluding much of the low-energy shelf. The near-flatness of the low-energy feature here is of great value in obtaining an accurate determination of A_T/A_G from the fits.

Figure 5 shows that the intensity ratio A_T/A_{TG} is indeed linearly dependent on μ , but that as observed by Nicholson *et al.*⁸ there is a finite intercept on the vertical axis. Figure 6 provides an even stronger indictment of the ICC model of Fig. 1(a) and Eqn (1); the deduced value of t_{ICC} exhibits a marked dependence on x-ray energy. It is obvious that such a dependence would not be observed in experiments limited to a narrow energy range. Finally, Fig. 7 shows that the photoelectron escape model of Shima *et al.* is also unable to explain the results.

We therefore propose a composite of the two models discussed; this is depicted in Fig. 2(c). An ICC layer is assumed in which every x-ray interaction results in a degraded event. Further degraded events arise from interactions in the subsequent region of complete charge collection if the photoelectron penetrates back to and creates some of its hole-electron pairs in the ICC layer.

Our approximations are to assume that all interactions occur in the silicon K shell, to neglect Auger electrons, to assume that the photoelectron energy is $E_x - 1.84$ keV and to assume a straight path for every such electron, with the range R given by Fitting's equation⁹

$$R = 0.09\rho^{-0.8}E^{1.3} \quad (10)$$

where ρ is the density of silicon and E the electron energy. The tail-to-peak ratio is then given by the equation

$$\frac{A_T}{A_G} = \frac{[1 - \exp(-\mu^*t_{ICC})] + I}{\exp(-\mu^*t_{ICC}) - I} \quad (11)$$

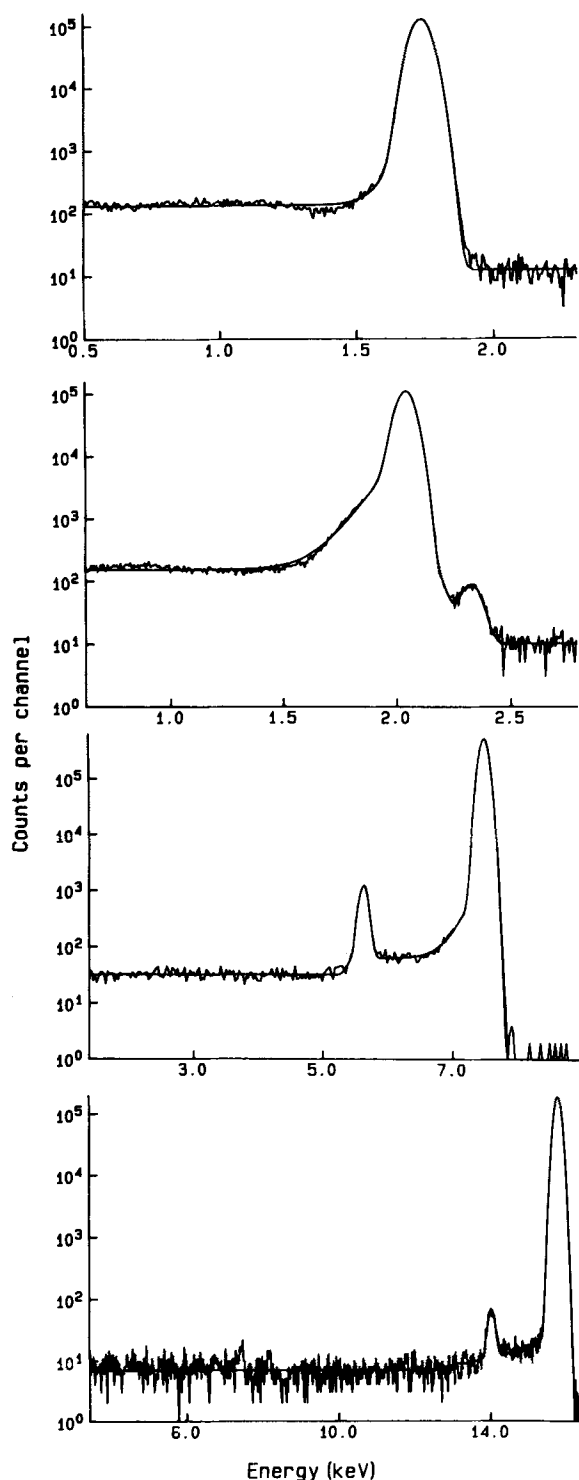


Figure 4. Fits of Eqn (6) to silicon $K\alpha_1$, zirconium $L\alpha_1$, nickel $K\alpha_1$ and zirconium $K\alpha_1$ lines. The smooth curve is the fit.

where

$$I = \int_{t_{\text{ICC}}}^{t_{\text{ICC}}+R} 0.5 \left(1 - \frac{z - t_{\text{ICC}}}{R} \right) \times \mu^* \exp(-\mu^*z) dz \quad (12)$$

Evaluation of this expression with a value $0.21 \mu\text{m}$ for t_{ICC} produced the upper curve in Fig. 7, which is in much better agreement with observation at photon energies up to 8 keV than the other two models. Such

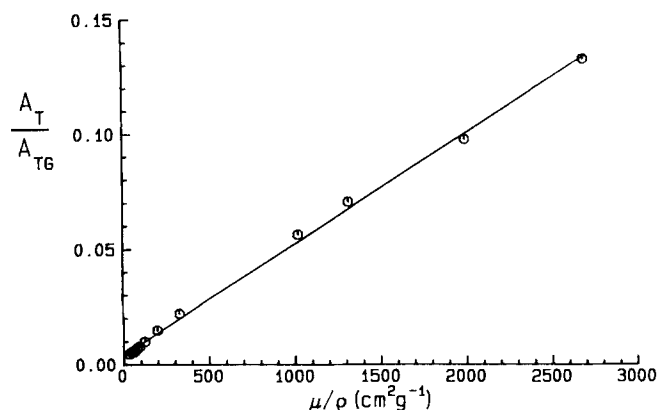


Figure 5. Fitted ratio of tail intensity to total intensity as a function of mass attenuation coefficient μ .

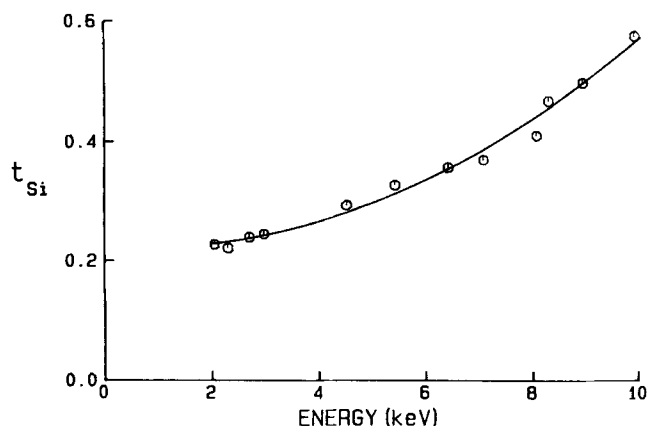


Figure 6. ICC layer thickness deduced from Eqn (1).

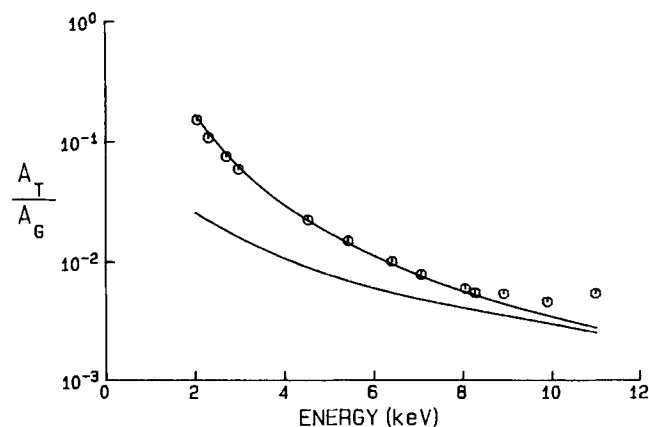


Figure 7. Comparison of measured tail-to-Gaussian intensity ratios with (a) Shima *et al.*'s model (lower curve) based on escape of Auger electrons and photoelectrons; (b) present calculation (upper curve) based on an ICC layer ($\eta < 1$) and including photoelectron escape from the $\eta = 1$ region.

agreement is not surprising when the photoelectron range is small compared with the thickness of the ICC layer. At higher x-ray energies where the range effects dominate the ICC effect, our predictions approach those of Shima *et al.*, and both underestimate the obser-

variations. At these energies, however, the observed tailing intensity is no longer decreasing as the x-ray energy increases, and other processes, e.g. inelastic scattering, are obviously coming into play.

This complete model is fairly successful. It should be emphasized, however, that although it provides an improvement over the previously used models it is still crude, especially in its assumption that the concept of electron range is useful at such low energies.

Finally, an attempt was made to obtain an upper limit for a true surface dead layer of silicon. Using the molybdenum $L\alpha_1$ x-ray spectrum with the criterion that the minimum detectable silicon thickness would result in a Si $K\alpha$ peak intensity equal to three standard deviations of the background in a region of width one FWHM centred at 1.74 keV, an upper limit of 0.002 μm was obtained for the thickness of this layer. If photoelectrons from the interaction were to penetrate to the region of non-zero charge collection, the Si $K\alpha$ peak would be smeared out towards higher energy and would be harder to discern. The maximum range of these electrons is 0.017 μm , so that such an effect is possible, but given that the ICC layer is about 0.2 μm thick, the effect is not likely to be large. One can conclude that any completely inactive layer is very much thinner than the ICC layer.

METALLIC CONTACT THICKNESS

We can proceed from the above to a useful determination of low-energy efficiency if the average thickness of the metal contact (t_c), nickel in this case, is also known. A caution is necessary at the outset, however: the term 'average thickness' is an idealization, since metal contacts are known to be non-uniform.

Customarily, t_c is measured by fluorescence. There have been many reports on measurement of gold contact thickness using the 14.4-keV γ rays of ^{57}Co , which can excite all three L subshells (binding energies 11.918, 13.733 and 14.353 keV). Maenhaut and Raemdonck¹ cautioned that this method is prone to error; using Mo $K\alpha$ x-rays of 17.4 keV to fluoresce the Au L lines they determined $t_c = 3.4$ nm, whereas the nominal value was 20 nm. They pointed out that in this case the gold L_3 photoelectrons have energy 5.5 keV, and thus have maximum range values of 0.1 μm in gold and 0.54 μm in silicon; they can easily penetrate the contact and ICC layers and deposit energy in the detector, causing a rightwards smearing of the Au $L\alpha$ line. This loss of fluorescence events into the continuum results in an erroneously low value for t_c .

For the present detector, whose contact is nickel (K binding energy 8.33 keV), t_c was measured by fluorescing successively with monoenergetic photons of energies 8.91, 9.89 and 10.98 keV; t_c is given by

$$\frac{A(\text{Ni } K\alpha)}{A(X)} = 0.5\mu^*t_c\omega_K p_\alpha \quad (13)$$

where the A s represent the peak intensities of Ni $K\alpha$ and the fluorescing line, μ^* the linear attenuation coefficient,¹² ω_K the fluorescence yield¹³ and p_α the $K\alpha$ frac-

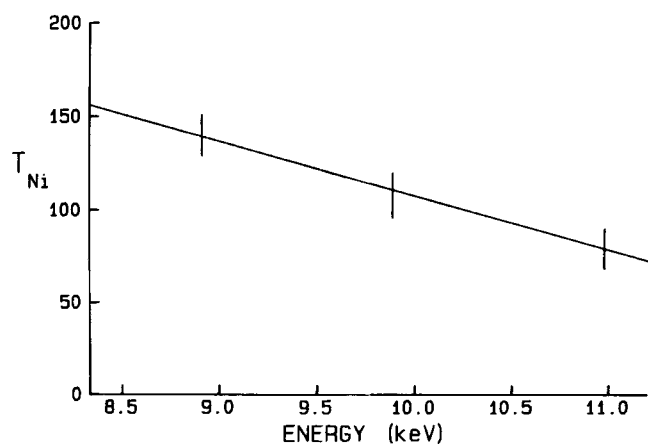


Figure 8. Measured thickness of the nickel contact as a function of the fluorescence energy. Each vertical bar represents the spread of ten successive measurements.

tion in the K x-ray series.¹⁴ The results (Fig. 8) show a decrease as the fluorescing x-ray energy increases, i.e. as more photoelectrons penetrate to the active silicon, thus supporting the argument of Maenhaut and Raemdonck. Using Cu $K\beta_1$ (8.91 keV) to fluoresce, the electron range is just 8 nm and the t_c values is 14 nm, so that here the effect will be small. Our preference is to extrapolate to zero photoelectron energy, producing the result $t_c = 15.5 \pm 0.5$ nm.

EFFICIENCY AT LOW ENERGY

For x-rays incident through a collimator that defines a fixed solid angle, the relative efficiency is usually quoted as

$$\varepsilon_1 = \exp[-(\mu_{\text{Be}}^* t_{\text{Be}} + \mu_c^* t_c + \mu_{\text{Si}}^* t_{\text{ICC}})] \quad (14)$$

where the μ^* s are linear attenuation coefficients.¹² The beryllium thickness t_{Be} may be taken as the nominal value or measured by weighing a duplicate, or preferably the quantity $\mu_{\text{Be}}^* t_{\text{Be}}$ may be determined directly by measuring the transmission of monoenergetic photons. The contact thickness t_c can be determined by fluorescence but, as is shown above, an accurate value will only result by measuring and removing the 'energy dependence' of t_c , or by fluorescing with an energy just above the contact binding energy. We have shown t_{ICC} is not well defined unless the modification of Eqn (11) is made. However, we prefer simply to replace the ICC term in the efficiency equation as follows:

$$\varepsilon_1 = \exp[-(\mu_{\text{Be}}^* t_{\text{Be}} + \mu_c^* t_c)][1 - (A_T/A_{\text{TG}})] \quad (15)$$

The component $S(i)$ is assumed to extrapolate horizontally to zero energy so that its area is just $H_s i_0$. Monte Carlo computations¹⁵ show that contributions to the tail from interactions in the contact are only a few percent and are confined to very low energies. A layer of ice is not included because this detector is equipped with a heating element for ice removal.

In evaluating Eqn (14) for ε_1 , the quantity A_T/A_{TG} shown in Fig. 9 was fitted as a log-log dependence on

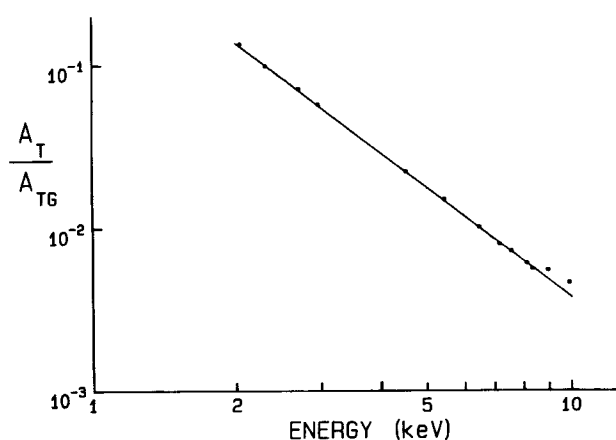


Figure 9. Energy dependence of the tail function A_T/A_{TG} .

x-ray energy, viz.

$$\ln A_T/A_{TG} = -(0.4525 + 2.2322 \ln E) \quad (16)$$

The absolute efficiency ϵ_A of the collimated detector was energies 5.414, 5.895 and 6.404 keV was measured using radionuclide standards of ^{54}Mn , ^{55}Fe and ^{57}Co , supplied by the Physikalische Technische Bundesanstalt. The ϵ_A/ϵ_i ratio averaged over these three energies was 0.00535 ($\pm 2\%$) and this ratio was used to convert the ϵ_i values to the absolute efficiency curve as shown in Fig. 10. We consider this curve to be accurate to better than 5%.

CONCLUSIONS

The main conclusion is that a simple extension of the widely used ICC model remedies the failure of that model to generate a unique thickness for the silicon layer of incomplete charge collection. It should be re-emphasized that even with this improvement the model remains crude.

The absolute efficiency of a Si(Li) detector is straightforward to determine at x-ray energies above 5 keV, thanks to the availability of radionuclide standards.² Below 5 keV there are essentially no standards and so a troublesome problem arises in this regime where the efficiency decreases rapidly with decreasing energy. Various techniques have been proposed, mostly involving x-rays excited in targets by electron, photon or proton bombardment.^{2,7} Some of these methods are indirect and some require equipment that is not readily accessible. Others⁷ involve the ICC model in Fig. 2(a), which has been shown here to be flawed.

The idea of deriving efficiencies from line shapes is not new, but past attempts have been frustrated by the inability to separate tailing events from the background.

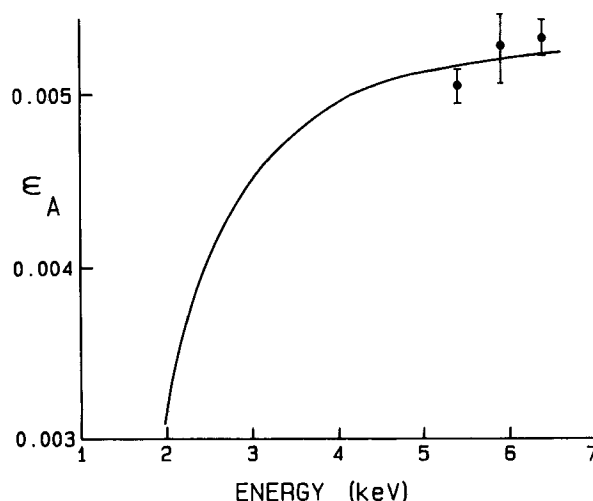


Figure 10. Absolute efficiency of collimated Si(Li) detector.

The background in this work, based on a monochromator is almost negligible, and the very uniform low-energy response of the detector results in an accurate determination of A_T/A_{TG} . However, attention must also be paid to the measurement of the metal contact's average thickness. We have shown that this can only be effected accurately if the x-rays used to fluoresce the contact have energy only very slightly in excess of the contact's atomic binding energy. The present results suggest that a significant under-estimate is likely in the widely used method of using a ^{57}Co radioactive source to measure the thicknesses of gold contacts.

We reiterate that this work is for a collimated detector. As Fig. 3 showed, A_T/A_{TG} rises towards the periphery owing to edge effects such as surface channels. One approach¹⁶ taken to describe the efficiency of an uncollimated detector is to assume the existence of a second ICC layer, this one having annular shape. Although this is operationally useful, it is probably no more valid than the concept of an ICC layer defined by Eqn (1), which we have shown here to be unsatisfactory.

Finally, we believe that line-shape measurements of the type described here, especially at low x-ray energies, provide a potentially useful tool for correlating the spectral response of a detector with details of the manufacturing technology. A study of the effects of variations in the fabrication process across a range of detectors might prove informative.

Acknowledgements

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